A conservative allocation completion for the VSL lane candidate

2026-10-21 — proposed adaptation; no model adoption or traffic experiment

Result and boundary. A small explicit allocator can satisfy common traffic and receiving budgets on the three-lane geometry. The choice between unsuccessful lane-change attempts continuing forward and committed directional fractions changes queue behavior. This note specifies the former and compares the latter. It establishes conditional algebraic admissibility, not a complete VSL/capacity-drop/task qualification or a calibrated lane-change law.

1. State, geometry and admissible inputs

Arrange lanes as E, A, B: ending lane, adjacent continuing lane, far continuing lane. Physical cell row i sends to row $i+1$ through a storage-free allocation node. An allowed movement goes straight or to an adjacent lane; it advances exactly one row. All boundary flows are evaluated from the old time level before any stock is updated. This diagonal transport is motivated by Carey et al. [1]; it replaces the principal source's same-cell lateral step followed by adjusted longitudinal flows [3].

For one boundary, let a index donor lanes and j receiver lanes. Use amounts per fixed time step, not rates:

$$d_a = \Delta t\, D_{i,a}, \qquad r_j = \Delta t\, R_{i+1,j}, \qquad 0 \le d_a \le n_{i,a}, \quad 0 \le r_j \le J_{i+1,j} - n_{i+1,j}. \tag{1}$$

n is continuous vehicle-equivalent stock, J jam storage, and D, R have units of vehicle-equivalents per unit time. Thus d, r, n, J have the same mass unit. Nothing is rounded. Source sending, receiving and any capacity-drop branches must actually satisfy (1); this is a condition, not a result of the allocator. Standard fixed-grid CTM bounds include

$$v_{\max}\Delta t/\Delta x \le 1, \qquad w_{\max}\Delta t/\Delta x \le 1. \tag{2}$$

For a triangular FD, $D \le v_{\max}k$ and $R \le w_{\max}(k^{\mathrm{jam}} - k)$ then imply the stock bounds. Modified source branches and the eventual Han-type sending restriction [4] need their own compatibility check; (2) alone does not certify every branch.

2. Proposed translation of incentives into requests

Let $P_{aj} \ge 0$ be the principal source's finite lateral preference for an allowed adjacent direction, before its disputed realization equations. Mask forbidden directions first. For a continuing donor define

$$s_a = \sum_{j \in \mathcal{L}_a} P_{aj}, \qquad b_{aj} = \frac{P_{aj}}{\max(1, s_a)} \quad (j \in \mathcal{L}_a),$$
$$b_{aa} = 1 - \sum_{j \in \mathcal{L}_a} b_{aj}, \qquad m_{aj} = d_a b_{aj}. \tag{3}$$

$\mathcal{L}_a$ is the set of allowed adjacent targets; $m_{aa} = d_a b_{aa}$. Hence $m_{aj} \ge 0$ and $\sum_j m_{aj} = d_a$. Existing valid fractions are unchanged; only a jointly oversubscribed donor is rescaled. This normalization is a proposed interface rule, not a native source equation. For an empty donor all requests are zero; no empty-density preference ratio is evaluated.

At the permanent end there is no E receiver: remove movements into it. Set $b_{EA} = 1$ for the last E donor. No $E \to B$ movement is permitted. Before the end, $E \to E$ remains a legal forward fallback even if its initial request happens to be zero. Fractions are evaluated anew from the next state; no persistent route/intention records are introduced.

3. Concrete candidate: allocate attempts, then allow forward fallback

First pass: share each receiving budget. For each target j, fix positive priorities p_{aj} for its nonzero request contributors. The proposal uses the donor's *reference geometric lane capacity* (times Δt, if a mass scale is desired), independently of advice and current demand. Equal reference capacities give equal priorities. This uses the capped fixed-priority sharing principle of Carey R3 and the MISO node literature [1, 2]; the three-lane composition is ours. Define

$$z_{aj} = \begin{cases} m_{aj}, & \sum_h m_{hj} \le r_j, \\ \min(m_{aj}, \lambda_j p_{aj}), & \sum_h m_{hj} > r_j, \end{cases} \quad \text{with} \quad \sum_h \min(m_{hj}, \lambda_j p_{hj}) = r_j. \tag{4}$$

No demand means zero flow. At zero supply take $\lambda_j = 0$. In the constrained positive case the flow vector is unique: increase the common allocation level until all the available receiving budget is used. A request that is fully served drops out of further sharing, leaving space for others. In particular, $0 \le z_{aj} \le m_{aj}$ and $\sum_a z_{aj} \le r_j$. A small or absent request cannot retain an unused reserved entitlement.

Second pass: use otherwise unused forward space. Define residual donor traffic and receiver space after (4):

$$e_a = d_a - \sum_j z_{aj} \ge 0, \qquad h_j = r_j - \sum_a z_{aj} \ge 0. \tag{5}$$

If the same-lane connection exists, send

$$y_a = \min(e_a, h_a), \qquad f_{aj} = z_{aj} + \mathbf{1}_{\{j=a\}} y_a. \tag{6}$$

When it does not exist, set $y_a = 0$. Each receiver has at most one such same-lane fallback donor. There is no second lateral reallocation and no same-step movement of newly arriving stock. The two passes decide one set of flows; they are not two physical time steps.

The elementary proposal gives one entering mass unit one receiving-budget unit. It sets Carey's extra headway multiplier to $\alpha = 1$ rather than adding another merging-loss law. This is an explicit modeling simplification. Extra gap consumption and slow-entry recovery are not thereby reproduced; the retained congestion/capacity-drop mechanism still needs qualification.

4. Strong rival: keep directional fractions as commitments

Instead feed d_a, b_{aj}, r_j and fixed input priorities into the published general junction allocation framework [2]. With full FIFO within an input lane, its directions are restricted together:

$$f_{aj} = b_{aj} F_a, \qquad 0 \le F_a \le d_a, \qquad \sum_a b_{aj} F_a \le r_j. \tag{7}$$

The source's priority/redistribution rule, not these constraints or maximum flow alone, completes the multi-input solution. This is the alternative's published algorithmic contract; it is not an arbitrary LP tie break. Blocked turns can hold back all directions from their input lane. No-FIFO instead caps each movement by its own m_{aj}; this still does not add the forward fallback of (6). These fractions need only be fixed within one evaluation, not for a driver's entire trip.

Both candidates require explicit normalized incentive and boundary mappings. Neither can silently import its source's general-node theorems into the full VSL plant. In particular the preferred candidate is *not* the Wright MIMO solver. Demand-proportional priorities have a source invariance warning [2, Remark 3.1]; using fixed priorities avoids that particular choice, but is not a proof of the preferred composition's nodal invariance.

5. Conditional admissibility proof for the preferred candidate

From (3)–(6),

$$f_{aj} \geq 0, \qquad \sum_j f_{aj} = \sum_j z_{aj} + y_a \leq d_a,$$
$$\sum_a f_{aj} = \sum_a z_{aj} + y_j \leq r_j. \tag{8}$$

For a receiver without a legal same-lane donor, its added y_j is zero. Thus every donor and receiver is shared once even with two lateral directions and simultaneous opposing changes. Individual desired fractions are not independent permissions to reuse mass. In the absence of lateral requests, the first pass gives the ordinary lane CTM flow $\min(d_a, r_a)$ and fallback adds zero.

Let $O_{i,a}$ sum all movements from cell (i, a) and $I_{i,a}$ sum all movements entering it from row $i-1$. The stock update is

$$n^+_{i,a} = n_{i,a} - O_{i,a} + I_{i,a}. \tag{9}$$

Because $O_{i,a} \leq d_{i,a} \leq n_{i,a}$ and $I_{i,a} \leq r_{i,a} \leq J_{i,a} - n_{i,a}$,

$$0 \leq n_{i,a} - d_{i,a} \leq n^+_{i,a} \leq n_{i,a} + r_{i,a} \leq J_{i,a}. \tag{10}$$

Every internal movement appears once as a departure and once as an arrival, so summing (9) conserves mass up to recorded external flows. Only allowed connections carry requests or fallback. Unserved final-E mass remains in the stock update; reaching the end is not an exit.

What replaces the earlier obstruction? There is no adjusted receiving term $R + L_{\text{in}} - L_{\text{out}}$ and no negative residual demand. Flows use nonnegative budgets, and realized outflow cannot exceed their shared donor. A departure's released storage becomes available through the next old-state evaluation. This changes the time/space stencil rather than asserting a sign correction to the native formula. The earlier source counterexamples remain unchanged; they are not relabeled as trajectories of this new model.

6. Local sensitivity and important missing guarantees

For a constrained receiver, fix the active sets H of fully served requests and U of requests limited by their priority entitlement. Then

$$\lambda_j = \frac{r_j - \sum_{a \in H} m_{aj}}{\sum_{a \in U} p_{aj}}, \qquad z_{aj} = \begin{cases} m_{aj}, & a \in H, \\ \lambda_j p_{aj}, & a \in U. \end{cases} \tag{11}$$

The denominator is positive. Hence the allocator is smooth on fixed positive request active sets wherever its input maps are smooth. Minima, the donor normalization threshold, empty requests and changes in these sets require branch-aware derivatives. This is a local algebraic observation, not finite difference validation, a mode-event treatment or useful-gradient evidence.

The source incentive and capacity-drop branches may themselves have nonsmooth or discontinuous transitions; no complete derivative statement is made about that joined model. Nor are continuum convergence, invariance or global throughput maximization claimed for this two-pass adaptation. The next page gives a concrete throughput limit. Admissibility alone does not establish a realistic merge service or a useful VSL task.

7. Hand cases: direct answers rather than a rollout

All numbers below are nonnegative allocator input amounts per step. They test the proposed budget rules, not the existence of a complete density, incentive and control trajectory with exactly these values.

Two neighbors overbook one donor. Set $d = 10$ and two lateral preferences $P_1 = P_2 = 7/10$. Equation (3) gives

$$(m_1, m_2, m_{\text{own}}) = (5, 5, 0). \tag{12}$$

With no other contenders and receiving budgets $(2, 8, 10)$, the first pass serves $(2, 5, 0)$. The remaining three use the own-lane connection:

$$(f_1, f_2, f_{\text{own}}) = (2, 5, 3), \qquad \sum_j f_j = 10. \tag{13}$$

No negative residual, mass multiplication or unused donor quota is hidden.

Failed attempt versus committed movement. Before the lane end let $d_E = 10$, $b_{EA} = b_{EE} = 1/2$, $r_A = 0$, $r_E = 10$, with no other contenders. Then

Rule	$E \to A$	$E \to E$
Attempts, then forward fallback	0	10
Fixed fractions, full FIFO	0	0
Fixed fractions, no FIFO, no fallback	0	5

At the permanent end remove $E \to E$ and set $b_{EA} = 1$: with $r_A = 0$ all ten remain in the last E cell for either main candidate.

Cooperation under constrained receiving. Let E and A each request 10 units into A, with equal positive reference priorities and $r_A = 8$. Equation (4) serves $(4, 4)$. If E requests only 1 and A still requests 10, it serves $(1, 7)$: unused entitlement is returned. This is fixed priority with demand caps, not a fixed realized merging fraction.

A real limitation: the preferred rule need not maximize total flow. Let $d_E = d_A = 10$, $b_{EA} = 4/5$, $b_{EE} = 1/5$, and $b_{AA} = 1$ before the lane end. Set $r_A = r_E = 10$ and use equal reference priorities. The first pass serves $E \to A = 5$, $A \to A = 5$, $E \to E = 2$; E's fallback adds 3. The final total is 15. If the eight lateral requests could instead all be waived before sharing, sending both complete donors forward would give 20. Our rule gives initial requests precedence; the later fallback does not revoke space they already claimed. A maximal total-flow objective with freely adjustable desires would be a different behavioral model.

Decision implication. The preferred rule offers a complete, deterministic allocation component with transparent limitations. Accepting its opportunity meaning and priority sequence is a substantive modeling choice. This proof does not choose it on the user's behalf or close the remaining task qualification.

References

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